Project Proposal

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Overview

Consider the following unconstrained non-linear optimization problem.

Problem 1 (Euler, 1740s)**.** *Given f, find the following minimizer $x(t)$:*

$$\min_{x(t)} \int_{t_1}^{t_2} f(t, x(t), x'(t))\ dt. \tag{1}$$

We will use the general problem above to specifically solve John Bernoulli's *brachistochrone ("least time") problem*, stated in modern terminology below.

Problem 2 (Brachistochrone, 1696)**.** *Solve Problem 1 for the specific integrand $f = f_0$ below:*

$$f_0(t, x(t), x'(t)) = \sqrt{\frac{1 + x'(t)^2}{2g \cdot x(t)}}.$$

We can solve this problem using a plethora of modern techniques that were not known at the time of its formulation; in particular, it was not known how to take the derivative of a function.

This project is composed of two parts based on the work of Lagrange and Euler: (1) derivation of the analytic solution $x(t)$, and (2) computational comparison against other elementary functions.

1 Analysis

This problem differs from others studied in this course in that we are asked to find a minimizing curve instead of a finite-dimensional vector. Lagrange followed Euler's work and rephrased the problem in a more recognizable form below:

$$\min_{\varepsilon} \int_{t_1}^{t_2} f(t, x(t) + \varepsilon \cdot y(t), x'(t) + \varepsilon \cdot y'(t))\ dt, \qquad y \in C([t_1, t_2)] \text{ such that } y(t_1) = y(t_2) = 0,$$

whose solution is $\varepsilon = 0$ iff $x(t)$ is a solution to the previous problem. This leads to a particularly nice Euler-Lagrange differential equation, which allows us to solve for x when given a particular f_0. We will carry out this derivation in explicit detail, which will look familiar as it was the historical basis for the method of Lagrange multipliers.

2 Computation

This problem is historically interesting because it is also the basis of calculus of variations. We will implement Euler's method of numerical integration and test it on other elementary curves (e.g. defined by the parabola, line, or Gaussian function) to prove that the analytically-dervied solution is optimal. Note that this numerical integration is easy when the curves being considered admit elementary anti-derivatives (as is the case for the parabola and the line) by application of the fundamental theorem of calculus, but we will see that Euler's method works in non-trivial cases when there is no elementary anti-derivative (as is the case for curves involving the Gaussian function, for example).

Time permitting, we may explore other analytic derivations or methods of numerical integration.